How much does the endpoint say about the switching rate?

Working note for direction S1 (moved out of Paper C, October 2026)

October 2026

1 Setting

This note was Section 8 and Appendix E of an earlier version of Paper C ("Face selection for slowly switching Markov chains and the planar persistent walk"). The mathematics is unchanged. It uses only Paper A [2], and this section recalls what it needs from there. Smiga, Radaelli, Binder and Landi [4] compute the Fisher information of the empirical distribution under a Gaussian approximation. Their leading term vanishes for the symmetric walk of Paper A, whose stationary law does not depend on the switching rate.

Kagey's board [1] has N steps. The first step is left or right with probability $\frac{1}{2}$, and each later step repeats the previous one with probability p. We write $q = 1 - p$ and $u = q/p$, the odds of a switch. We write $L = \log N$, and $T = Nq$ is the expected number of switches (up to the first step). $P_N(k)$ is the probability of k right steps, that is, that the ball lands in bin k. The endpoints have $P_N(0) = P_N(N) = p^{N-1}/2$.

For $a, b \geq 1$ let $S_{a,b}(u) = \sum_w u^{s(w)}$, the sum over the words w with a right steps and b left steps, where $s(w)$ is the number of switches of w. Then $S_{a,b}(q/p)$ is the ratio $P_N(a)/P_N(0)$ of the probabilities of bin a and bin 0 after $a + b = N$ steps (the run count of [2]). Let $S_N = S_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$. It has nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N, and $p_N = 1/(1 + u_N)$ is the crossing of the middle and the endpoints. Put $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$. Paper A proves that $z_N = L - \frac{1}{2}\log L + c_0 + O(\log L/L)$. We call this the second-order expansion of z_N.

2 The information in the endpoint

This note asks a statistical question about Kagey's board. If we only see the bin where the ball lands, how much do we learn about the switching probability? The answer is that the bin keeps about a fraction $1/(2T)$ of the Fisher information of the whole path, so about $1/(2\log N)$ at Kagey's crossing. We use the notation of Section 1. The first step is left or right with probability $\frac{1}{2}$ each, and each later step repeats the one before it with probability $p = 1 - q$, where $0 < q < 1$, and $T = Nq$. Put $\rho = 1 - 2q$, $u = q/p$ and $y = Nu = T/p$. Let J be the number of switches, K the number of right steps, $X = 2K - N$ the endpoint and $W = X/N$. A path with J switches has probability $\frac{1}{2}p^{N-1-J}q^J$. So $J \sim \mathrm{Bin}(N-1, q)$, and the path carries the Fisher information $(N-1)/(pq)$ about q. The endpoint carries $I_K(q) = \sum_{k=0}^{N} (\partial_q \mathbb{P}(K = k))^2/\mathbb{P}(K = k)$, and we call $e_N(q) = I_K(q)\, pq/(N-1)$ the *efficiency* of the endpoint. It is the fraction of the information left when we see only the endpoint.

The endpoint sees the switches only through how far the walk spreads. A normal X with mean 0 and variance Np/q, close to $\operatorname{Var} X$ for large T, would carry the information $\frac{1}{2}(\partial_q \log(Np/q))^2 = 1/(2p^2q^2)$, which is the fraction $1/(2(N-1)pq) \approx 1/(2T)$. Theorem 2.3 confirms this.

Proposition 2.1 (a correlation ratio)**.** *For $N \geq 2$ and $0 < q < 1$,*

$$e_N(q) = \frac{\operatorname{Var}(\mathbb{E}[J \mid K])}{\operatorname{Var} J}.$$

For $0 < k < N$ we have $\mathbb{E}[J \mid K = k] = \frac{d}{dt} \log S_{k,N-k}(e^t)$ at $t = \log(q/p)$, where $S_{a,b}$ is the polynomial of Section 1. For $k \in \{0, N\}$ we have $J = 0$.

Proof. The score of a path is $(J - (N-1)q)/(pq)$. Since $\mathbb{P}(K = k)$ is a sum of path probabilities, $\partial_q \log \mathbb{P}(K = k) = \mathbb{E}[(J - (N-1)q)/(pq) \mid K = k]$. As $\mathbb{E}J = (N-1)q$, this gives $I_K(q) = \operatorname{Var}(\mathbb{E}[J \mid K])/(pq)^2$, and we divide by $(N-1)/(pq) = \operatorname{Var} J/(pq)^2$. By the run count of [2], for $0 < k < N$ the probability $\mathbb{P}(K = k, J = j)$ is $\frac{1}{2}p^{N-1}u^j$ times the number of words with k right steps, $N-k$ left steps and j switches. Summing over j gives $\mathbb{P}(K = k) = \frac{1}{2}p^{N-1}S_{k,N-k}(u)$, so $\mathbb{E}[J \mid K = k] = uS'_{k,N-k}(u)/S_{k,N-k}(u)$. Only the two constant paths have $K \in \{0, N\}$. □

This identity, checked in Lean, says that e_N is the correlation ratio of J on K. Since $\mathbb{E}[J \mid K]$ is the orthogonal projection of J onto the functions of K in L^2, the Cauchy–Schwarz inequality gives

$$e_N(q) \geq \frac{\operatorname{Cov}(J, g(K))^2}{\operatorname{Var} J \operatorname{Var} g(K)} \qquad \text{whenever } \operatorname{Var} g(K) > 0. \tag{1}$$

Proposition 2.2 (few switches)**.** *(a) For fixed $N \geq 3$, $e_N(q) = 1 - \frac{N-2}{2}q + O_N(q^2)$ as $q \to 0$.*

(b) Let $q \to 0$ with $T = Nq$ bounded. Then $e_N(q) \to 1$ if and only if $T \to 0$.

In words, the endpoint loses information as soon as paths with two switches appear. The reason is that one switch and two switches both leave the endpoint uniform on the interior bins, so an interior bin cannot tell them apart.

Proof. We measure the loss $1 - e_N$ by how uncertain J still is once we know the bin. By Proposition 2.1 and the decomposition of the variance,

$$1 - e_N = \frac{\mathbb{E} \operatorname{Var}(J \mid K)}{\operatorname{Var} J}, \qquad \operatorname{Var} J = (N-1)pq.$$

On the two end bins $J = 0$, so only the interior bins count. Fix an interior bin $0 < k < N$. Exactly two words with k right steps have one switch, and exactly $N-2$ have two switches ($k-1$ that start to the right and $N-k-1$ that start to the left). Each word with j switches has probability $\frac{1}{2}p^{N-1-j}q^j$. So

$$\mathbb{P}(J = 1, K = k) = p^{N-2}q, \qquad \mathbb{P}(J = 2, K = k) = \frac{N-2}{2}\, p^{N-3}q^2.$$

(a) Here N is fixed and $q \to 0$. By these two formulas, given $K = k$ the number of switches is 2 with probability $\frac{N-2}{2}q + O_N(q^2)$ and larger than 2 with probability $O_N(q^2)$. Otherwise it is 1. So J given $K = k$ is close to a coin with a small chance of the value 2, and $\operatorname{Var}(J \mid K = k) = \frac{N-2}{2}q + O_N(q^2)$. The interior bins have total probability $1 - p^{N-1} = (N-1)q + O_N(q^2)$. Hence $\mathbb{E} \operatorname{Var}(J \mid K) = \frac{(N-1)(N-2)}{2}q^2 + O_N(q^3)$, and dividing by $(N-1)pq$ gives the claim.

(b) We prove an upper and a lower bound for $1 - e_N$ that hold for all $N \geq 2$ and $0 < q < 1$.

Upper bound. The conditional mean $\mathbb{E}[J \mid K]$ is the function of K closest to J in mean square. So replacing it by another function of K, namely the indicator of an interior bin, can only raise the error. Note that $J \geq 1$ exactly on the interior bins, so J minus this indicator is 0 when $J = 0$ and $J - 1$ otherwise. Since $(J-1)^2 \leq J(J-1)$ for $J \geq 1$,

$$\mathbb{E}\operatorname{Var}(J \mid K) \leq \mathbb{E}\big(J - \mathbf{1}\{0 < K < N\}\big)^2 \leq \mathbb{E}[J(J-1)] = (N-1)(N-2)q^2.$$

(The last step is the second factorial moment of $J \sim \operatorname{Bin}(N-1, q)$.) Dividing by $(N-1)pq$ gives $1 - e_N \leq (N-2)q/p \leq T/p$. This tends to 0 if $T \to 0$.

Lower bound. We use only the values 1 and 2 of J. If a variable equals 1 with probability a and 2 with probability b (and anything else otherwise), then its variance is at least ab. (The variance is half the mean of $(J_1 - J_2)^2$ for two independent copies, and with probability $2ab$ one copy is 1 and the other is 2.) So

$$\mathbb{E}\operatorname{Var}(J \mid K) \geq \sum_{0<k<N} \frac{\mathbb{P}(J=1, K=k)\mathbb{P}(J=2, K=k)}{\mathbb{P}(K=k)}.$$

By the Cauchy–Schwarz inequality and $\sum_{0<k<N} \mathbb{P}(K=k) \leq 1$, the right side is at least

$$\Big(\sum_{0<k<N} \sqrt{\mathbb{P}(J=1, K=k)\mathbb{P}(J=2, K=k)}\Big)^2 = \frac{(N-1)^2(N-2)}{2}\, p^{2N-5}q^3,$$

where we used the two formulas above and that there are $N-1$ interior bins. Dividing by $(N-1)pq$ gives

$$1 - e_N \geq \frac{(N-1)(N-2)}{2}\, p^{2N-6}q^2.$$

Conclusion. Let $q \to 0$ with T bounded. If $T \to 0$, the upper bound gives $e_N \to 1$. If T does not tend to 0, then $T \geq t_0 > 0$ along a subsequence, and $T \leq t_1$ for some t_1. Along it $N \to \infty$, so $(N-1)(N-2)q^2 = T^2(1 + o(1))$ and $p^{2N-6} = e^{-2T}(1 + o(1))$. So the lower bound is $\frac{1}{2}T^2e^{-2T} + o(1) \geq \frac{1}{2}t_0^2e^{-2t_1} + o(1)$, and e_N does not tend to 1. □

Theorem 2.3 (many switches)**.** *Let $R_N^2(q) = \operatorname{Cov}(J, X^2)^2/(\operatorname{Var} J \operatorname{Var} X^2)$ be the squared correlation of J and X^2. For $N \geq 2$ and $0 < q < 1$,*

$$R_N^2(q) = \frac{1}{2(N-1)pq} \cdot \frac{\big[1 + \rho^N - p(1-\rho^N)/T\big]^2}{V},$$

where

$$V = 1 - \frac{1 + 8\rho + \rho^2 + 8\rho^{N+1}}{4pT} + \frac{\rho(1-\rho^N)\big[2(1+2\rho)(2+\rho) - \rho(1-\rho^N)\big]}{8p^2T^2}.$$

There are absolute constants C and N_0 such that for $N \geq N_0$, $0 < q \leq \frac{1}{2}$ and $1 \leq T \leq N^{1/4}$,

$$0 \leq e_N(q) - R_N^2(q) \leq C\Big(\frac{1}{T^3} + \frac{T^3}{N}\Big).$$

Consequently, in the same range and with an absolute implied constant,

$$e_N(q) = \frac{1}{2T} + \frac{1}{4T^2} + O\Big(\frac{1}{T^3} + \frac{T^3}{N}\Big).$$

In words, when T is large but at most $N^{1/4}$, the endpoint keeps about the fraction $1/(2T)$ of the information, and almost all of what it keeps is carried by the single statistic X^2. The lower bound $e_N \geq R_N^2$ is (1) with $g(K) = X^2$, for all $N \geq 2$ and $0 < q < 1$. For the upper bound, Proposition 2.1 and the uniform Bessel estimate for every bin in [2] (recalled in (2) below) make $\mathbb{E}[J \mid K]$ close to $y\sqrt{1-W^2} + \frac{1}{2}$. As W is of order $T^{-1/2}$, this is close to $y(1 - W^2/2) + \frac{1}{2}$, which lies in the span of 1 and X^2. Appendix A proves the closed form and the bound.

Proof of the expansion in Theorem 2.3. If $T \leq 18$, then $|e_N - \frac{1}{2T} - \frac{1}{4T^2}| \leq 1 \leq 18^3 T^{-3}$, since e_N and $\frac{1}{2T} + \frac{1}{4T^2}$ lie in $[0, 1]$. Let $T \geq 18$. As $q \leq \frac{1}{2}$, we have $p \geq \frac{1}{2}$ and $0 \leq \rho < 1$, and $\rho^N \leq e^{-2Nq} = e^{-2T} \leq T^{-2}$. The middle term of V lies in $[0, 9/T]$, and since $1 + 8\rho + \rho^2 = 10 - 20q + 4q^2$ it equals $\frac{5}{2T} + O(q/T + T^{-2})$. The last term lies in $[0, 9/T^2]$. So $V \geq \frac{1}{2}$, $1/V = 1 + \frac{5}{2T} + O(q/T + T^{-2})$, the squared bracket is $1 - \frac{2}{T} + O(q/T + T^{-2})$, and $1/(2(N-1)pq) = (1 + O(q + 1/N))/(2T)$. Multiplying, $R_N^2 = \frac{1}{2T} + \frac{1}{4T^2} + O(T^{-3} + q/T + 1/(NT))$ with absolute constants. Since $q/T = 1/N \leq T^3/N$, the bound on $e_N - R_N^2$ gives the expansion. □

Corollary 2.4 (at Kagey's crossing)**.** *Let p_N be the value of p at which the middle bin and the endpoints are equally likely, and let $q_N = 1 - p_N$ and $z_N = Nq_N$ (Section 1). Then*

$$e_N(q_N) = \frac{1}{2L} + \frac{\log L + 1 + \log(8/\pi)}{4L^2} + O\Big(\frac{(\log L)^2}{L^3}\Big).$$

Proof. Since $S_N(u) \geq 2u$, we have $u_N \leq \frac{1}{2}$, so $p_N \geq \frac{2}{3}$ and $q_N \leq \frac{1}{3}$. By the second-order expansion of z_N in [2], $z_N = L(1-\varepsilon)$ with $\varepsilon = (\frac{1}{2}\log L - c_0)/L + O(\log L/L^2)$. So Theorem 2.3 applies with $T = z_N$ for large N and gives $e_N(q_N) = \frac{1}{2z_N} + \frac{1}{4z_N^2} + O(L^{-3})$. Here $\frac{1}{2z_N} = \frac{1}{2L} + \frac{\log L - 2c_0}{4L^2} + O((\log L)^2/L^3)$, $\frac{1}{4z_N^2} = \frac{1}{4L^2} + O(\log L/L^3)$ and $-2c_0 = \log(8/\pi)$. □

So $e_N \sim 1/(2\log N)$ there, but the relative correction is of order $(\log\log N)/\log N$ and is large at every N we computed. The product $2Le_N$ is 1.74 at $N = 100$ and 1.35 at $N = 3200$ (Table 1).

N	z_N	$e_N(q_N)$	$2z_N e_N$	$2Le_N$	$R_N^2(q_N)$	$\frac{1}{2z_N} + \frac{1}{4z_N^2}$
50	2.8998	0.24117	1.3987	1.8869	0.20873	0.20216
100	3.5034	0.18869	1.3221	1.7379	0.16642	0.16309
400	4.7501	0.12681	1.2047	1.5196	0.11715	0.11634
1600	6.0246	0.09433	1.1366	1.3919	0.09007	0.08988
3200	6.6684	0.08361	1.1151	1.3496	0.08070	0.08060

Table 1: The efficiency of the endpoint at Kagey's crossing $q_N = 1 - p_N$, computed from the exact law, with $z_N = Nq_N$ and $L = \log N$. The last two columns are the squared correlation of Theorem 2.3 and its expansion to two terms.

Many balls

Now drop n independent balls and record only the histogram of their bins. How well can q be estimated? By the Cramér–Rao bound no unbiased estimator has variance below $1/(nI_K(q))$, and the maximum likelihood estimator attains this as $n \to \infty$. The next corollary says that a much simpler estimator does almost as well. Put $m(q) = \mathbb{E}_q X^2$, a polynomial in q.

Corollary 2.5 (the moment estimator)**.** *Let $X_1, \dots, X_n$ be the endpoints of n independent balls with switching probability q, and suppose that $R_N^2(q) > 0$. Then $m'(q) \neq 0$, and with probability tending to 1 the equation $m(\hat{q}) = \frac{1}{n}\sum_{i=1}^n X_i^2$ has a solution $\hat{q}_n$ that converges to q in probability. For this solution*

$$\sqrt{n}\,(\hat{q}_n - q) \Rightarrow N\Big(0, \frac{pq}{(N-1)R_N^2(q)}\Big) \qquad (n \to \infty).$$

So its asymptotic efficiency, relative to the bound $1/(nI_K(q))$, is $R_N^2(q)/e_N(q)$. There is an absolute constant C' such that, with N_0 as in Theorem 2.3, for $N \geq N_0$, $0 < q \leq \frac{1}{2}$ and $1 \leq T \leq N^{1/4}$,

$$1 - C'\Big(\frac{1}{T^2} + \frac{T^4}{N}\Big) \leq \frac{R_N^2(q)}{e_N(q)} \leq 1.$$

In words, among all estimators that see only the bins, matching the mean square of the endpoint loses at most a fraction of order $1/T^2 + T^4/N$ of the information. At Kagey's crossing the efficiency R_N^2/e_N is 0.87 at $N = 50$ and 0.97 at $N = 3200$ (Table 1).

Proof. This is the usual argument for a method of moments estimator. The one thing special to our model is that the slope $m'(q)$ is a covariance with J, and this is what brings in R_N^2.

The slope. The probability of a path is a polynomial in q, and its score is $(J - (N-1)q)/(pq)$. So $m'(q) = \mathbb{E}[X^2(J - (N-1)q)]/(pq) = \operatorname{Cov}(J, X^2)/(pq)$. This is not zero because $R_N^2(q) > 0$.

The estimator. Since $m'(q) \neq 0$, there is an interval U around q on which m is strictly monotone, with a continuous inverse on $m(U)$. By the law of large numbers the mean of the X_i^2 tends to $m(q)$ in probability. So with probability tending to 1 it lies in $m(U)$, and then we let $\hat{q}_n$ be the solution in U. It converges to q because the inverse is continuous.

The limit law. By the central limit theorem, $\sqrt{n}$ times the difference of the mean of the X_i^2 and $m(q)$ tends to $N(0, \operatorname{Var} X^2)$. The delta method divides the variance by $m'(q)^2$. So the limit variance is

$$\frac{\operatorname{Var} X^2}{m'(q)^2} = \frac{(pq)^2 \operatorname{Var} X^2}{\operatorname{Cov}(J, X^2)^2} = \frac{(pq)^2}{R_N^2(q) \operatorname{Var} J} = \frac{pq}{(N-1)R_N^2(q)},$$

where we used the definition of R_N^2 and $\operatorname{Var} J = (N-1)pq$.

The efficiency. Since $I_K(q) = e_N(q)(N-1)/(pq)$, the ratio of $1/I_K(q)$ to this variance is R_N^2/e_N. It is at most 1 by (1) with $g(K) = X^2$.

The lower bound. Put $\epsilon = T^{-2} + T^4/N$. Theorem 2.3 gives $e_N - R_N^2 \leq C\epsilon/T$. Let C_1 be the constant in its expansion, so that $e_N \geq \frac{1}{2T} - C_1\epsilon/T$, and put $C' = 4\max(C, C_1)$. If $C'\epsilon \geq 1$ the claim is trivial, since $R_N^2/e_N \geq 0$. Otherwise $C_1\epsilon < \frac{1}{4}$, so $e_N \geq \frac{1}{4T}$. Then

$$1 - \frac{R_N^2}{e_N} = \frac{e_N - R_N^2}{e_N} \leq \frac{C\epsilon}{T} \cdot 4T = 4C\epsilon \leq C'\epsilon.$$

□

A Proof of the Fisher information theorem

We keep the notation of Section 2. Let $\sigma_i \in \{-1, 1\}$ be the direction of step i. Then $X = \sigma_1 + \cdots + \sigma_N$, and J counts the $i < N$ with $\sigma_{i+1} \neq \sigma_i$. Put $Z = \mathbb{E}[J \mid K]$, so $e_N = \operatorname{Var} Z / \operatorname{Var} J$ by Proposition 2.1.

Lemma A.1 (exact moments)**.** *For $N \ge 1$ and $0 < q < 1$,*

$$\begin{aligned}
\mathbb{E}X^2 &= \frac{Np}{q} - \frac{\rho(1-\rho^N)}{2q^2}, \\
\operatorname{Cov}(J, X^2) &= -\frac{p}{q^2}\left[Nq(1+\rho^N) - p(1-\rho^N)\right], \\
\mathbb{E}X^4 &= \frac{3N^2p^2}{q^2} - \frac{Np(1+10\rho+\rho^2)}{2q^3} + \frac{\rho(1+2\rho)(2+\rho)(1-\rho^N)}{2q^4} - \frac{3Np\rho^{N+1}}{q^3},
\end{aligned}$$

and $\operatorname{Var} X^2 = 2N^2p^2V/q^2$*, with V as in Theorem 2.3.*

Proof. We show that each moment, as a sequence in N, is a combination of the sequences N^i and $N^i\rho^N$ with $i \le 4$. Two such combinations that agree at ten values of N agree everywhere, and a script compares them at those values exactly.

Let $m_N(\phi, \chi) = \mathbb{E}e^{\phi X + \chi J}$. With $A_{ab} = \mathbb{P}(\sigma_2 = b \mid \sigma_1 = a)\, e^{\phi b + \chi \mathbf{1}[a \ne b]}$ and $v_a = \frac{1}{2}e^{\phi a}$ we have $m_N = v^\top A^{N-1}\mathbf{1}$. So $\sum_{N\ge1} m_N z^N = U/D$ as formal power series, where $U(z) = zv^\top \operatorname{adj}(I - zA)\mathbf{1}$ and $D(z) = \det(I - zA)$ have degree at most 2 in z. For a partial derivative ∂ of order m in (ϕ, χ), induction and the quotient rule give $\partial(U/D) = U_\partial/D^{m+1}$ with $\deg U_\partial \le 2m+2$, as each derivative multiplies the numerator by D or a derivative of D. At $\phi = \chi = 0$ the matrix A has trace $1+\rho$ and determinant ρ, so $D(z) = (1-z)(1-\rho z)$. Let $q \ne \frac{1}{2}$, so its roots 1 and $1/\rho$ are distinct. By partial fractions U_∂/D^{m+1} is a constant plus a combination of $(1-z)^{-j}$ and $(1-\rho z)^{-j}$ with $j \le m+1$, whose coefficients of z^N are $\binom{N+j-1}{j-1}$ and $\binom{N+j-1}{j-1}\rho^N$. So for $N \ge 1$ the sequence ∂m_N at 0 lies in $\mathcal{S}_m = \operatorname{span}\{N^i, N^i\rho^N : i \le m\}$. A sequence in $\mathcal{S}_m$ satisfies the recurrence with characteristic polynomial $(t-1)^{m+1}(t-\rho)^{m+1}$, so it is 0 if it vanishes at $N = 1, \dots, 2m+2$.

The moments $\mathbb{E}X^2$, $\mathbb{E}[JX^2]$ and $\mathbb{E}X^4$ are derivatives of m_N of order at most 4 at 0, so they lie in $\mathcal{S}_4$. The claimed formulas for $\mathbb{E}X^2$, $\mathbb{E}X^4$ and $\mathbb{E}[JX^2] = \operatorname{Cov}(J, X^2) + (N-1)q\,\mathbb{E}X^2$ lie in $\mathcal{S}_2$. So it is enough to check them at $N = 1, \dots, 10$. The script `paperC/verify_sticky_fisher.py` computes the three moments exactly for $N \le 12$ by a dynamic program with q as a symbol, and every difference is 0. For fixed N both sides are rational in q with no pole at $\frac{1}{2}$, so the case $q = \frac{1}{2}$ follows. Finally, with R standing for ρ^N and ρR for ρ^{N+1}, the claim $\mathbb{E}X^4 - (\mathbb{E}X^2)^2 = 2N^2p^2V/q^2$ is an identity of rational functions of N, q and R, which the same script checks with sympy. □

Proof of the closed form in Theorem 2.3. For $N \ge 2$, X^2 is N^2 on the constant paths and $(N-2)^2$ on the paths that switch only at the last step, so $\operatorname{Var} X^2 > 0$ and $V > 0$ by Lemma A.1, which also gives $\operatorname{Cov}(J, X^2) = -\frac{Np}{q}[1 + \rho^N - p(1-\rho^N)/T]$, so $\operatorname{Cov}(J, X^2)^2/\operatorname{Var} X^2$ is the squared bracket over $2V$. We divide by $\operatorname{Var} J = (N-1)pq$. □

Lemma A.2 (convexity)**.** *Let f and g be convex and differentiable on $[t-\delta, t+\delta]$, with $0 \le g - f \le \eta$ and $g'' \le \kappa$ there. Then $|f'(t) - g'(t)| \le \eta/\delta + \kappa\delta/2$.*

Proof. By convexity and Taylor's theorem, $f'(t) \le (f(t+\delta) - f(t))/\delta \le (g(t+\delta) - g(t) + \eta)/\delta \le g'(t) + \kappa\delta/2 + \eta/\delta$, and the lower bound is the same with $t - \delta$. □

Lemma A.3 (a Bessel ratio)**.** *For $t > 0$ let $R(t) = I_1(t)/I_0(t)$. Then $(\sqrt{1+t^2} - 1)/t \le R(t) < 1$. In particular $0 < 1 - R(t) \le 1/t$.*

Proof. Since $I_0' = I_1$ and $I_1'(t) = I_0(t) - I_1(t)/t$, we have $R' = 1 - R/t - R^2$, and $R(t) = t/2 + O(t^3)$ at 0. At a first point t_0 with $R(t_0) = 1$ we would have $R'(t_0) \ge 0$, but the equation gives $-1/t_0$. So $R < 1$. Put $s = \sqrt{1+t^2}$ and $\ell = (s-1)/t$. Then $1 - \ell/t - \ell^2 = \ell/t = 1/(s+1)$ and $\ell' = 1/(s(s+1))$, so $\ell' \le 1 - \ell/t - \ell^2$. For $d = R - \ell$ this gives $d' \ge -d/t - (R^2 - \ell^2) = -d(1/t + R + \ell)$. So Θd is

nondecreasing, where $\Theta(t) = t\exp\int_0^t(R+\ell)$. As $\ell = t/2 + O(t^3)$, we have $d = O(t^3)$ and $\Theta d \to 0$ at 0. Hence $d \ge 0$. Finally $1 - \ell = (t + 1 - s)/t \le 1/t$. □

Lemma A.4 (moments of the endpoint)**.** *(a) For each $m \ge 1$ there is C_m with $\mathbb{E}W^{2m} \le C_m T^{-m}$ for all $0 < q < 1$ and $T \ge 1$.*

(b) If $0 < q \le \frac{1}{2}$, then $\mathbb{E}W^4 \le 3p^2/T^2 + 9/(2T^4)$.

Proof. (a) Add a step σ_{N+1} to the walk. Since $\mathbb{E}[\sigma_{i+1} \mid \sigma_{\le i}] = \rho\sigma_i$, the $\xi_{i+1} = \sigma_{i+1} - \rho\sigma_i$ are martingale differences. Summing $\sigma_{i+1} - \sigma_i = -2q\sigma_i + \xi_{i+1}$ over $i \le N$ gives $X = (\sigma_1 - \sigma_{N+1} + \Xi)/(2q)$ with $\Xi = \sum_{i=2}^{N+1}\xi_i$. Also ξ_{i+1}^2 is $4q^2$ if step $i+1$ repeats step i and $4p^2 \le 4$ otherwise. So $\sum \xi_i^2 \le 4q^2N + 4J'$, where $J' \sim \mathrm{Bin}(N, q)$ counts the switches among the $N+1$ steps. Burkholder's inequality [3, Ch. 2] gives $\mathbb{E}\,\Xi^{2m} \le c_m\mathbb{E}(4q^2N + 4J')^m \le c_m 8^m((q^2N)^m + \mathbb{E}J'^m)$, and $q^2N \le T$. The falling factorial moments of J' are at most T^j, and J'^m is a combination of the falling factorials of order $j \le m$ with nonnegative coefficients. So $\mathbb{E}J'^m \le b_m T^m$ for $T \ge 1$. As $|\sigma_1 - \sigma_{N+1}| \le 2$, we get $\mathbb{E}X^{2m} \le (2q)^{-2m}2^{2m-1}(2^{2m} + \mathbb{E}\,\Xi^{2m}) \le C_m q^{-2m}T^m$, and $\mathbb{E}W^{2m} = N^{-2m}\mathbb{E}X^{2m} \le C_m T^{-m}$.

(b) We divide the formula for $\mathbb{E}X^4$ by N^4. As $0 \le \rho < 1$, the two terms of order T^{-3} are at most 0 and the one of order T^{-4} is at most $9/(2T^4)$. □

Proof of the bound in Theorem 2.3. Let $\mathcal{P}$ be the orthogonal projection in L^2 onto the span of 1 and X^2. Since X^2 is a function of K, $\mathrm{Cov}(Z, X^2) = \mathrm{Cov}(J, X^2)$. So $\mathrm{Var}\,Z = \mathrm{Cov}(J, X^2)^2/\mathrm{Var}\,X^2 + \mathbb{E}(Z - \mathcal{P}Z)^2$, and $e_N - R_N^2 = \mathbb{E}(Z - \mathcal{P}Z)^2/\mathrm{Var}\,J \ge 0$. Also $\mathbb{E}(Z - \mathcal{P}Z)^2 \le \mathbb{E}(Z - G)^2$ for $G = y(1 - W^2/2) + \frac{1}{2}$. From now on $0 < q \le \frac{1}{2}$ and $1 \le T \le N^{1/4}$, and C is an absolute constant that may change at each use. Then $p \ge \frac{1}{2}$, $T \le y \le 2T$ and $\mathrm{Var}\,J = (N-1)pq \ge T/4$. It suffices to show $\mathbb{E}(Z - G)^2 \le C(T^{-2} + T^4/N)$.

The plan is as follows. By Proposition 2.1, Z is the logarithmic derivative of the polynomial $S_{k,N-k}$ at u. Paper A gives a Bessel function that is close to this polynomial. Closeness of two functions does not give closeness of their derivatives in general, but it does for convex functions, and this is Step 1. Step 2 computes the logarithmic derivative of the Bessel function and finds $y\sqrt{1 - W^2} + \frac{1}{2}$ up to a small error. Step 3 replaces the square root by $1 - W^2/2$. Steps 4 and 5 add up the errors in mean square.

For $0 < k < N$ put $s_k = \sqrt{k(N-k)}$, $c_k = N/(2s_k)$ and $w_k = 2s_k u$. On $K = k$ we have $s_k^2 = N^2(1 - W^2)/4$, so $c_k = (1 - W^2)^{-1/2} \ge 1$, $w_k = y\sqrt{1 - W^2}$ and $c_k w_k = y$. Let $\mathcal{I}_k(z) = I_0(2z) + c_k I_1(2z)$ and $\widetilde{S}_k(v) = 2v\,\mathcal{I}_k(s_k v)$, which approximates $S_{k,N-k}(v)$. Here I_0 and I_1 are the modified Bessel functions. By the uniform Bessel estimate for every bin in [2] (the theorem "uniform estimate for every bin" there, with $a = k$ and $b = N - k$),

$$0 \le 1 - \frac{S_{k,N-k}(v)}{\widetilde{S}_k(v)} \le \frac{Nv^2}{2} + v \qquad \text{for all } v > 0. \tag{2}$$

Let $\Lambda_k(t) = \log S_{k,N-k}(e^t)$ and $\beta_k(t) = \log\widetilde{S}_k(e^t)$. By Proposition 2.1, $Z = \Lambda_K'(\log u)$ on $0 < K < N$.

Step 1. Both Λ_k and β_k are convex, as logarithms of series in e^t with nonnegative coefficients. Let $|t - \log u| \le 1$. Then $v = e^t \le eu$ and $y \le 2N^{1/4}$ give $Nv^2/2 + v \le (e^2y^2/2 + ey)/N \le (2e^2 + 2e)/\sqrt{N} \le \frac{1}{2}$ for $N \ge 1700$. As $-\log(1 - \varepsilon) \le 2\varepsilon$ for $0 \le \varepsilon \le \frac{1}{2}$, (2) gives $0 \le \beta_k - \Lambda_k \le \eta$ there, with $\eta = (e^2y^2 + 2ey)/N$. Next, $\beta_k(t) - t - \log 2$ is the logarithm of $\mathcal{I}_k(s_k e^t) = \sum_n a_n e^{nt}$ with $a_n \ge 0$. So β_k'' is the variance of n under the weights $a_n e^{nt}$, at most the mean of n^2. As $(z\frac{d}{dz})^2$

maps $I_0(2z)$ to $4z^2 I_0(2z)$ and $I_1(2z)$ to $(4z^2+1)I_1(2z)$, this mean is at most $4z^2+1 \le e^2y^2+1 = \kappa$, since $z = s_k e^t \le ey/2$. For $N \ge 4$, $\delta = \sqrt{2\eta/\kappa} \le 1$ and Lemma A.2 gives

$$\left|\Lambda_k'(\log u) - \beta_k'(\log u)\right| \le \sqrt{2\eta\kappa} \le \frac{2e^2y^2 + 2ey + 1}{\sqrt{2N}} \le \frac{11(1+y)^2}{\sqrt{N}}. \tag{3}$$

Step 2. Write $z = s_k e^t$, $w = 2z$ and $R = R(w)$. By the formulas for I_0' and I_1',

$$\beta_k'(t) = 1 + \frac{z\mathcal{I}_k'(z)}{\mathcal{I}_k(z)} = 1 + \frac{wR + c_k w - c_k R}{1 + c_k R} = w + \frac{1 + w(c_k - 1)(1-R)}{1 + c_k R}.$$

At $t = \log u$ we have $w = w_k$, so $\beta_k'(\log u) = w_k + B_k$ with B_k the last fraction. By Lemma A.3, $0 < R < 1$ and $w(1-R) \le 1$. So the numerator of B_k lies in $[1, c_k]$ and is at most $1 + wc_k = 1 + y$. Hence $0 < B_k \le 1 + y$. Subtracting $\frac{1}{2}$ from $1/(1+c_kR) \le B_k \le c_k/(1+c_kR)$ gives

$$-\frac{c_k - 1}{2} \le B_k - \frac{1}{2} \le \frac{(c_k - 1) + c_k(1-R)}{2(1+c_kR)} \le \frac{c_k - 1}{2} + \frac{c_k^2}{2y},$$

by $c_k(1-R) \le c_k/w_k = c_k^2/y$. If $W^2 \le \frac{1}{2}$ on $K = k$, then $c_k^2 \le 2$ and $c_k - 1 \le W^2$, since $(1-v)^{-1/2} - 1$ is convex, vanishes at 0 and is below $\frac{1}{2}$ at $v = \frac{1}{2}$. So $|B_k - \frac{1}{2}| \le W^2/2 + 1/y$ there.

Step 3. Let $\psi(a) = 1 - a^2/2 - \sqrt{1-a^2}$ for $|a| \le 1$. The series of $\sqrt{1-v}$ is $1 - v/2 - \sum_{j\ge 2} b_j v^j$ with $b_j \ge 0$ and $\sum_j b_j = \frac{1}{2}$, so $0 \le \psi(a) \le a^4/2$. On $K = k$, $w_k = y(1 - W^2/2) - y\psi(W)$. So on $0 < K < N$ we have $Z - G = (Z - \beta_K'(\log u)) + (B_K - \frac{1}{2}) - y\psi(W)$, and on the atoms $K \in \{0, N\}$ we have $Z = 0$, $W^2 = 1$ and $Z - G = -(y+1)/2$.

Step 4. By Lemma A.4 and Markov's inequality, $\mathbb{E}W^4 \le 8/T^2$, $\mathbb{E}W^8 \le CT^{-4}$ and $\mathbb{P}(W^2 > \frac{1}{2}) \le 16\,\mathbb{E}W^8 \le CT^{-4}$. Also $\mathbb{P}(K \in \{0, N\}) = p^{N-1} \le e^{-q(N-1)} \le e^{1/2 - T}$.

Step 5. By Step 3 and Minkowski's inequality, $\|Z - G\|_2$ is at most the sum of the L^2 norms of the three terms of Step 3 on $0 < K < N$ and of $\frac{y+1}{2}$ on the atoms. By (3) the first is at most $99T^2/\sqrt{N}$, as $(1+y)^2 \le 9T^2$. For the second we use Step 2 where $W^2 \le \frac{1}{2}$ and $|B_K - \frac{1}{2}| \le 1 + y$ elsewhere, so its square is at most $\mathbb{E}W^4/2 + 2/y^2 + C(1+y)^2T^{-4} \le C/T^2$. The third is at most $\frac{y}{2}(\mathbb{E}W^8)^{1/2} \le C/T$, and the last is $\frac{y+1}{2}\mathbb{P}(K \in \{0,N\})^{1/2} \le CTe^{-T/2} \le C/T$. So $\mathbb{E}(Z-G)^2 \le C(T^4/N + T^{-2})$, and dividing by $\operatorname{Var} J \ge T/4$ gives the bound. Only Step 1 used $N \ge 1700$, so $N_0 = 1700$ works. □

References

[1] P. Kagey, *Problem 131*, Open Problem Collection, https://peterkagey.com/problems/131/ (accessed October 2026).

[2] A. Pemmasani, *Uniform Bessel bounds for endpoint crossings in the persistent random walk*, preprint, 2026. arXiv:2610.06935.

[3] P. Hall and C. C. Heyde, *Martingale Limit Theory and Its Application*, Academic Press, New York, 1980.

[4] J. A. Smiga, M. Radaelli, F. C. Binder and G. T. Landi, *Stochastic metrology and the empirical distribution*, Phys. Rev. Research **5** (2023), 033150. https://doi.org/10.1103/PhysRevResearch.5.033150.